

KAIZEN:PROBLEM STATEMENTS

Hackathon Rules & Guidelines

EVENT TIMELINE

- Start: February 21st, 11:00 AM IST
- End: February 22nd, 10:00 AM IST
- Duration: 24 hours exactly
- Submission Deadline: February 22nd, 10:00 AM IST (no late submissions)

CORE RULES

1. 24-Hour Build Only
 - All coding must happen between start/end times
 - No pre-written code, templates, or components allowed
 - Working prototypes started before 11 AM Feb 21st = immediate disqualification
2. GitHub Submission Only
 - Use GitHub repository provided by coordinators (no personal repos)
 - Include `README.md` with:
 - Problem statement chosen
 - Live demo link (Vercel/Netlify/ Railway)
 - Team name
3. No External APIs will be provided

MISCONDUCT = IMMEDIATE DISQUALIFICATION

- Pre-written code or templates
- Late submissions (after 10 AM Feb 22nd)

SUPPORT & CLARIFICATIONS

- Mentors for clarifications
- Problem statement questions: Ask mentors directly
- Technical help: Mentors can guide approach, not code

JUDGING CRITERIA

Judged purely on:

1. Innovation - Novel approach to the problem
2. Creativity - Unique solution design
3. Approach - Technical solution quality
4. Uniqueness - Originality of implementation
5. Presentation - Demo clarity and delivery

Problem Statements

CATEGORY 1: FINOPS & COST OPTIMIZATION

Problem 1: Idle Resource Auto-Scheduler

Problem Statement:

Cloud teams waste money running idle dev/test VMs during off-hours. Build a system that detects underutilized resources and proposes/executes shutdown schedules to cut costs.

Challenge Question:

How can you identify idle resources, create automatic shutdown schedules, and prove monthly cost savings?

Expected Outcomes:

1. Resource dashboard showing idle detection + cost metrics
 2. Scheduling UI with preset shutdown/startup times
 3. Simulated action log proving savings estimate
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Problem 2: Daily Budget Watchdog

Problem Statement:

Finance teams need real-time alerts when spending trends will exceed budget by month-end. Catch overspend early, not at bill time.

Challenge Question:

How can you forecast end-of-month spend, trigger alerts at threshold breach, and recommend one cost-saving action?

Expected Outcomes:

1. Dashboard with current spend + month-end forecast
 2. Alert UI showing threshold breach + recommendation
 3. Simulated action (e.g., scale down instance)
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CATEGORY 2: REVOPS & LEAD MANAGEMENT

Problem 3: Instant Lead Triage & Sales Handoff

Problem Statement:

Qualified leads get lost in manual triage. Automate lead enrichment, qualification, and routing to the right sales queue.

Challenge Question:

How can you capture, enrich, and route qualified leads automatically without manual handoff delays?

Expected Outcomes:

1. Lead capture form + enrichment display
 2. Qualification rules engine + routing logic
 3. Queue assignment confirmation + handoff simulation
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Problem 4: Personalized Follow-Up Generator

Problem Statement:

Sales reps waste time writing generic follow-ups. Generate personalized email/SMS/LinkedIn options they can edit and send quickly.

Challenge Question:

How can you generate contextual follow-ups fast enough that reps prefer it to writing from scratch?

Expected Outcomes:

1. Lead profile + 2+ message templates (email, SMS)
 2. Highlighted personalization slots + edit UI
 3. Send simulation with confirmation
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CATEGORY 3: SOCIAL IMPACT & ACCESSIBILITY

Problem 5: Adaptive Micro-Lesson Generator

Problem Statement:

Learners with disabilities need 10-minute focused lessons in multiple formats (audio, large text, high-contrast). Generic lectures don't work.

Challenge Question:

How can you generate accessible micro-lessons with multiple consumption options and a quick knowledge check?

Expected Outcomes:

1. Topic input → lesson page (goal + steps + example)
 2. 2+ accessibility toggles (large text, audio, high-contrast)
 3. 3-question quiz with instant feedback
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CATEGORY 4: ENTERPRISE PRODUCTIVITY & AUTOMATION

Problem 6: Support Ticket Classifier & Escalation

Problem Statement:

Support queues are chaotic—urgent issues wait, simple questions get mislabeled. Auto-classify tickets, route to right team, escalate critical ones.

Challenge Question:

How can you automatically triage tickets by urgency and category, then escalate critical issues to managers?

Expected Outcomes:

1. Ticket intake form + classification engine (urgency + category)
 2. Routing logic to 3+ teams + escalation rule trigger
 3. SLA assignment + queue confirmation
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Problem 7: Expense Report Validator

Problem Statement:

Finance spends hours checking expense compliance. Automate policy checks, flag violations, and route approvable reports for fast payment.

Challenge Question:

How can you validate expenses against policy rules, explain violations, and reduce review time by 50%+?

Expected Outcomes:

1. Expense entry form + policy rules engine (3+ rules)
 2. Violation detection with explanation + correction suggestions
 3. Approval/rejection recommendation + reimbursement calculation
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Problem 8: Candidate Screening & Interview Scheduler

Problem Statement:

Recruiting is slow because screening and scheduling are manual. Score candidates against job requirements and auto-schedule interviews.

Challenge Question:

How can you score candidate fit, rank them, and schedule top candidates with zero back-and-forth?

Expected Outcomes:

1. Job requirements + candidate profile intake (5+ samples)
2. Fit scoring (0–100) + ranking + auto-reject logic
3. Interview scheduler UI with proposed times + booking confirmation